multimedia

MDPI

Review

Development of a Generative AI-Based Workflow for the Design and Integration of 3D Assets in XR Environments for Research

José Luis Rubio Tamayo [1,*] and Mary Anahí Serna Bernal [2]

[1] Faculty of Communication, URJC, 28942 Madrid, Spain
[2] Faculty of Arts and Design, Universidad Nacional Autónoma de México, Mexico City 04510, Mexico; anahi_serna@comunidad.unam.mx
* Correspondence: jose.rubio.tamayo@urjc.es

Abstract

Scalable production of interactive 3D assets is a key requirement for XR-based applications, yet the functional integration of GenAI-generated assets into game engines remains challenging for non-expert users. This article proposes and validates a Prompt-to-Trigger workflow that links GenAI-based asset ideation and generation with the implementation of basic interactive behaviors (triggers) in accessible XR platforms. The study adopted a qualitative and exploratory approach, using systematic observation throughout a two-stage development process. This process included an initial phase where 3D assets were generated and refined using tools such as Tripo AI and Meshy, followed by an optimization stage to ensure compatibility with Blender and XR environments like A-Frame and Godot, and subsequently, the creation of AI-powered activation scripts. The results show that GenAI's current 3D outputs frequently exhibit topological inconsistencies and rigging errors that compromise performance and real-time interoperability, requiring cleanup and optimization before deployment. The Prompt-to-Trigger workflow formalizes this bridge, positioning AI assistance as a functional layer for iterative logic generation. The resulting model provides non-expert creators with structured, actionable framework to prototype complex XR experiences for applied domains like education and multimedia communication.

Keywords: generative artificial intelligence; extended reality; virtual reality; 3D assets

1. Introduction

Extended reality (XR) has increased the capacity for developing production models by integrating generative artificial intelligence (GenAI) into workflows [1], but XR content production remains a highly complex process. This is especially relevant since virtual environments facilitate dynamic simulations and skill acquisition [2].

XR technologies continue to evolve, making it necessary to systematically study how they function in practice and what conditions allow for optimizing production flows and the integration of their components. Lee [3] proposes a unified conceptual model of immersive experience, emphasizing the importance of anchoring XR design decisions in structured dimensions rather than relying on ad hoc practices. In parallel, Ratican et al. [4] describe a meta-reality development pipeline that links generative AI models with XR content production, signaling a shift towards workflow-driven methodologies, reflecting an active but still unconsolidated research landscape [5–8].

This study examines the multidisciplinary production work related to XR and its multiple phases and skill sets, from the initial creation of 3D assets to the implementation

Check for updates

Academic Editor: Jun Kong

Received: 27 November 2025
Revised: 27 March 2026
Accepted: 30 March 2026
Published: 7 April 2026

Copyright: © 2026 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license.

of interaction logic through triggers. These requirements justify a more granular, phase-by-phase view of XR production and reinforce the need for workflow models and standards that improve consistency and reproducibility. In practice, modeling or scanning assets is only an initial step; much of the effort lies in making assets usable in real time. This includes defining interaction behavior and completing production prep such as rigging, animation and material/UV setup. Workflow accounts such as Rizvic et al. [9] document this structure in applied VR production and show how interactive events and immersive storytelling depend on tightly coupled artistic and technical decisions.

While XR adoption in learning-oriented contexts has increased, the broader challenge concerns the feasibility of producing high-quality interactive 3D content across applied domains. Specifically, GenAI-generated assets often exhibit low-quality topology that fails to meet standard geometric quality indicators necessary for simulation and rendering [10]. Furthermore, issues such as UV/texturing artifacts and limited editability arise from automated unwrapping methods that lack semantic awareness, producing fragmented maps incompatible with professional workflows [11]; even advanced implicit models frequently result in outputs that look rough or lack fine details required for high-fidelity immersion [12]. Beyond asset-level quality, Tu et al. [13] note that XR development remains difficult to modify for domain specialists who are not programmers, because small changes often require tool-specific scripting and interaction artifacts, allowing non-expert users to adjust application behavior by modifying control parameters rather than authoring triggers or engine-level logic. These coupled constraints explain why existing contributions that focus on isolated steps (generation or scripting) are insufficient and motivate an end-to-end workflow that makes the handoffs explicit.

The overall objective of this research is to propose a global and accessible Prompt-to-Trigger workflow that supports non-expert creators in designing, preparing, and implementing interactive 3D assets for XR in applied multimedia contexts using GenAI tools. Specifically, the study pursues the following three objectives:

- Examine the asset-generation phase by evaluating GenAI 3D outputs under different input conditions in terms of mesh quality, coherence, and texturing readiness.
- Analyze the integration phase by detailing the practical challenges of importing and preparing these assets for accessible implementation environments.
- Conceptualize the interaction phase by specifying how GenAI assistance can support trigger implementation as the step that transforms a prepared asset into a functional interactive component. The remainder of this paper presents the methodology, the main contributions and discussion of the proposed workflow.

2. Literature Review: GenAI and XR Content Creation

2.1. XR Context and Adoption (Applied Domains)

The development of immersive communication positions XR as a fundamental concept for understanding the ecosystem of information and communication technologies. As authors such as Qayyum et al. [14] point out, "the XR concept functions as an umbrella term that encompasses Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR)," among others. Other authors such as Memarian and Doleck [15] reaffirm this idea, describing it as a broad term that covers reality across a spectrum. Alkaeed et al. [16], for their part, describe it as a totally immersive digital universe and a virtual world that combines physical reality with digital information, seeking to converge the phenomena of artificial intelligence, on the one hand, and XR, on the other. Together, these definitions show that XR is used both as a taxonomic umbrella (VR/AR/MR) and as a continuum-based socio-technical space where AI convergence becomes relevant.

This potential has led researchers such as Guo et al. [17] and Rangel-de Lázaro and Duart [18] who document a growing consensus on the potential of XR in various applied domains. From this applied perspective, Zwoliński et al. [19] analyze how XR and related metaverse developments are influencing institutional adoption processes and adoption at the organizational level. From the standpoint of XR–AI integration, Barbu et al. [20] refer to this convergence as an approach that can enable immersive and interactive experiences beyond traditional methods.

2.2. Barriers and the Production Bottleneck

Despite this potential, the high cost of devices limits the large-scale adoption and implementation of XR in creative contexts, affecting its feasibility and scalability [21]. This leads designers to favor more affordable configurations; moreover, high implementation costs are identified by Akhtar and Rawol [22] as a major challenge, and they emphasize affordability as a key condition for enabling broader access. In the same direction, Mande and Bhaskar [23] identify high implementation costs and the lack of technical infrastructure as significant barriers that limit the widespread adoption of immersive technologies. Reinforcing this view at the review level, Yan, Li, and Lowell [24] categorize infrastructure and cost constraints as a recurring issue in AI–XR systems, noting that hardware requirements can be inaccessible for many institutions. These findings suggest that the economic constraints are structural rather than marginal; they fundamentally dictate the viability of XR configurations and, by extension, necessitate workflows that prioritize accessibility alongside lightweight deployment strategies.

Kawka et al. [25] identify the scarcity of high-quality XR content that enables ad hoc implementation as an equally critical barrier hindering adoption, beyond the constraints of hardware and infrastructure. This deficit responds fundamentally to a "production bottleneck". As Yosun Chang [26] notes, traditional 3D software utilizes interfaces with steep learning curves, creating a complex environment for audiences unfamiliar with design—a limitation reinforced by Sahebnasi et al. [27] who describe these tools as inaccessible to non-experts. This phenomenon persists today; generative AI has not yet fully substituted human processes but has instead opened research lines that, while technically advanced, make the development ecosystem extremely complex, as reflected in the studies of Rubio-Tamayo et al. [28]. Furthermore, Sahebnasi et al. [27] highlight the significant time and effort required for production as a decisive factor that complicates the adaptation of XR content to specific training lines.

This production bottleneck has driven the search for systems that democratize 3D content creation in XR, accelerating the emergence of workflows that connect generation with deployment. However, relying on external repositories is not a scalable solution for interactive XR pipelines, because assets often enter the workflow as static objects and lose functional properties required for interaction. Hunde and Woldeyohannes [29] note that kinematic relationships defined in CAD environments may not be preserved after import into VR platforms, and they further highlight meshing as one of the most time-consuming and error-prone stages—both of which reinforce the downstream workload required to make assets usable.

Beyond asset preparation, the literature increasingly points to investigating integrated workflows that make the transition from content models to XR-ready implementations more seamless. Fakhimi [30] cites GenXR as an efficiency-driven approach to accelerated model transfer into XR, but they also note that many proposals still do not cover the full process, including evaluation, redesign, and the iterative improvement that AI–XR integration can enable. In the same direction, Vallasciani et al. [31] argue that many XR systems restrict users to predetermined actions and interactions; they propose AI-supported platforms that

allow users to collaborative personalization and the generation of new elements, helping advance more accessible XR production processes.

2.3. GenAI + 3D Workflows and the Gap Addressed by Prompt-to-Trigger

Kawka et al. [25] focus on XR readiness and propose mechanisms for generating 3D models optimized for real-time pipelines, emphasizing very specific constraints such as size reduction and level of detail (LOD). On the other hand, Sahebnasi et al. [27] show that 3D generation with GenAI (e.g., Shap-E) can indeed be brought into engines like Unity or WebXR, although they acknowledge that programming remains an obstacle for non-expert creators. This gap is also recognized by Sun [32], who describes virtual production as having a steep learning curve due to the need for technical skills in various areas, and by Constantinescu et al. [33] who point out that world-building workflows can become cumbersome even when using accessible tools. In the ideation stage, Mendoza García et al. [34] reinforce the value of prompt-driven conceptualization to accelerate the transition from ideation to integrable results. Taken together, these contributions suggest that it is still rare to find approaches that connect ideation, asset preparation, and implementable interactivity within a complete and accessible flow; this is precisely where Prompt-to-Trigger's proposal comes in.

Beyond XR-oriented discussions, several studies examine GenAI-assisted 3D workflows in manufacturing, CAD, product design, and materials engineering, proposing optimization strategies and staged process models for domain-specific production pipelines [35–38]. While these contributions demonstrate the broader relevance of integrating AI with 3D design, they primarily address industrial objectives rather than the constraints of interactive XR deployment. This contrast reinforces the need for XR-specific workflows that remain feasible for non-expert creators and explicitly support the transition from generated content to interactive, deployable assets.

Regarding the application of artificial intelligence, specifically the application of generative artificial intelligence to 3D design in the specific context of virtual environment development, this line of work has been theorized since the late 1990s and early 2000s, shaping both the research trajectory and the technological evolution observed today. While it is true that 3D design and components related to graphic design form two of the fundamental pillars of virtual reality environment development, in recent years this XR medium has increasingly required hybridization with other design areas, such as space design, product design [39,40] experience design, or even architecture [41,42], a discipline where potential convergence already existed in the 1990s [43] and where virtual reality was already defined as "the ultimate representation" applied to this specific context.

This evolution over the years situates current XR production as a convergent medium between design practice and media art, highlighting its relevance in digital media production [44]. This has led authors such as Gong [45] to point out how AI and XR converge by intertwining 3D creation and design. This convergence helps explain why XR creation involves cross-domain competencies and why workflow-oriented approaches are necessary to make production feasible for non-expert creators.

Given the challenges posed by this type of production, the recent literature frames GenAI as a disruptive factor by redefining creative workflows within digital and multimedia production, including XR processes [34]. In generative design, Peckham et al. [46] describe them as a gradual shift away from exclusively manual making toward processes that combine automation, support for ideation, and workflow reconfiguration across different methods and use cases. This transition is related to GenAI's capacity to produce new content patterns from data [47]. In design work, that capability matters because it changes how exploration happens; prompting becomes a way to test and refine ideas. Li et al. [48]

report how UX design professionals perceive GenAI as a tool that reshapes creative work through prompt-based interaction and iterative generation. These perspectives justify treating GenAI not as a single tool, but as an enabling layer that reorganizes how ideas are externalized, iterated, and translated into producible assets—an assumption that grounds the workflow focus of this study. While this technological expansion has reached three-dimensional production through specialized models for rapid asset creation and integration into engines like Blender or Unreal Engine, empirical accounts warn that the integration remains incomplete and still relies on substantial technical steps and expertise [38].

Although the academic literature often addresses either asset generation or technical integration, research on an end-to-end workflow that remains accessible to non-expert creators is still scarce. To address this gap, the article presents the Prompt-to-Trigger workflow, which links prompt-driven ideation and asset creation with the subsequent implementation of basic interactive behavior. The operational definitions and workflow stages are specified in the following sections, where the model is detailed and evaluated in relation to XR production constraints.

Prior work usually improves a single segment of the XR production chain; while some studies prioritize rapid asset generation and visual ideation, others focus on technical preparation for real-time performance. A third line of work treats interaction as a standard scripting step and leaves outside the workflow discussion. However, the boundary conditions between these stages are rarely made explicit, so it is difficult to determine when GenAI outputs stop being assets and when they become implementable interactive components.

The Prompt-to-Trigger workflow is designed around those handoffs; it connects prompt-driven ideation, 3D generation, required post-processing, platform integration, and trigger implementation as successive stages with different roles and constraints. By making the transition explicit, the manuscript's contribution is framed in terms of operational continuity rather than isolated capabilities.

Figure 1 provides a conceptual framing of the study by positioning the proposed workflow at the intersection of the following three domains: extended reality, generative AI, and 3D content creation. The diagram is used to define the scope of the manuscript and to make explicit that the contribution does not belong exclusively to any single domain. Instead, the workflow is presented as a bridging approach that connects current GenAI capabilities with the technical and integration requirements of XR environments.

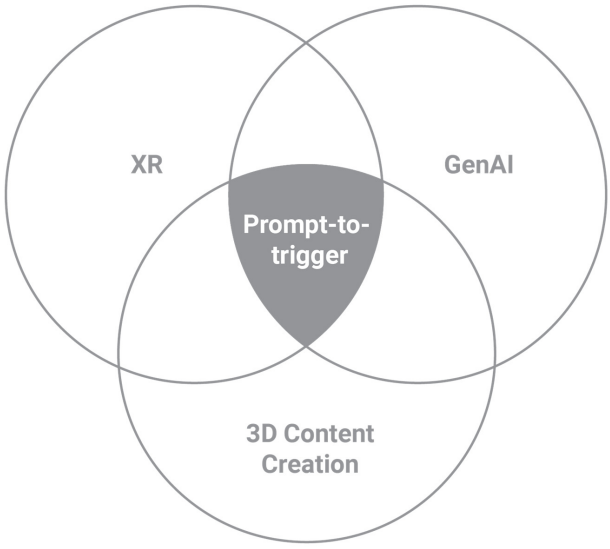


Figure 1. Euler diagram of XR, generative AI, and 3D content creation showing the location of the proposed Prompt-to-Trigger workflow. Source: authors' elaboration.

Within this framing, the XR domain is understood as a context that imposes practical constraints on how 3D assets are experienced and deployed, including spatial conditions shaped by tracking and by the physical environment, as well as real-time performance requirements. The generative AI domain refers to the recent capacity to produce 3D assets through prompt-driven systems, which can accelerate early prototyping but often requires additional preparation before use in real-time applications. The 3D content creation domain represents established production practices needed to deliver functional assets that can be optimized, imported, and used reliably in XR. The overlap highlighted in the diagram therefore motivates the Prompt-to-Trigger workflow as a structured pathway that links asset generation to production-readiness and, finally, to interactive integration.

3. Materials and Methods

This article adopts a qualitative, reflective, and exploratory approach, structured as an empirical study based on systematic observation and pattern identification within creative workflows. The study also includes a narrative literature review to contextualize GenAI-based 3D asset creation and XR integration and to inform the structure of the proposed Prompt-to-Trigger workflow. To ensure methodological rigor, the research design is grounded in the Augmented Artificial Intelligence conceptual framework [49]. Adhering to this model, the designer's role is not defined as a constant guide, but as the final auditor, responsible for identifying and correcting qualitative errors and context-specific deviations that the generative agents cannot detect autonomously.

3.1. Research Hypotheses

To guide the exploratory analysis and define the scope of the Prompt-to-Trigger model, the following three core hypotheses were formulated regarding the feasibility of the workflow:

H1: *Current generative AI technologies enable non-expert users to generate initial 3D objects in a simple and intuitive way.*

H2: *The optimization of a hybrid workflow allows for the efficient integration of generated 3D assets into XR environments.*

H3: *AI-assisted script generation (triggers) can reduce the technical barriers for non-expert creators to design functional interactive behaviors in XR.*

3.2. Methodological Design and Procedure

The methodology follows a structured pipeline in Figure 2, which proceeds from a narrative literature review to the identification of technical patterns, and finally to the induction of the workflow models. This design aligns with the sociotechnical perspective of "Implementing and Scaling AI" [50], which describes the shift from early validation to operational deployment when scaling a workflow for immersive environments.

Following the empirical methodology described by Begemann and Hutson [51], a hybrid workflow was established for this study. The specific tools selected, along with their function and analysis parameters, are detailed in Table 1. The procedure employed a digital simulation strategy [52] that applied these tools sequentially across the following three phases:

1. Ideation and reference: text-to-image generation to produce visual concepts.
2. 3D objects synthesis: generation of 3D geometry.
3. Optimization and integration: required post-processing before engine integration.

Table 1. Key tools used in the exploratory workflow and evaluation criteria. Source: authors' elaboration from compilation of different sources and software testing.

Tool	Purpose in Workflow	Stage	Evaluation Criteria (Codes)
Large Language Models (LLMs)	Support prompt iteration, documentation, and decision traceability across steps	Ideation/Planning	C, T
Tripo AI (web platform; Holymolly Ltd., Hong Kong, China; accessed on 21 July 2025)	Generate fast 3D drafts from text/image to accelerate prototyping	3D Draft	Q, I, T
Tencent (GenAI platform)	Alternative generative platform used for comparison and variation during exploration	Ideation/Asset Draft	Q, C, T
Meshy (web platform; Meshy LLC, Santa Clara, CA, USA; accessed on 21 July 2025)	Generate/texture 3D assets and iterate variations for props and scene elements	3D Draft/Texturing	Q, C, I, T
Luma AI (web platform; Luma AI, Inc., Palo Alto, CA, USA; accessed on 22 July 2025)	Capture/derive 3D assets (e.g., reconstruction) as input for the pipeline	3D Acquisition/Draft	Q, I, T
Spline (web platform; Spline, Inc., Middletown, DE, USA; accessed on 22 July 2025)	Quick 3D layout/blocking and interactive scene mockups (early spatial planning)	Spatial Blockout	C, T, I
Blender (version 4.2; Blender Foundation, Amsterdam, The Netherlands)	Clean-up and preparation: retopology, UVs, optimization, export-ready assets	Refinement/Optimization	Q, I, T
Unity (version 2022.3.62f3; Unity Technologies, San Francisco, CA, USA) /WebXR Device API (W3C standard)	Assemble scenes, implement interactions, and test performance in real-time deployment contexts	Integration/Deployment	Q, I, T
ElevenLabs (web platform; Eleven Labs Inc., 23 July 2025) + Suno (web platform; Suno, Inc., Cambridge, MA, USA; accessed on 23 July 2025) (complementary)	Voice and music assets for multimedia narrative context (not central object of analysis)	Narrative Support	Q, T

Codes: Q = output quality (mesh/texture integrity), C = controllability and repeatability, I = interoperability (standard formats/pipeline compatibility), T = time and clean-up effort.

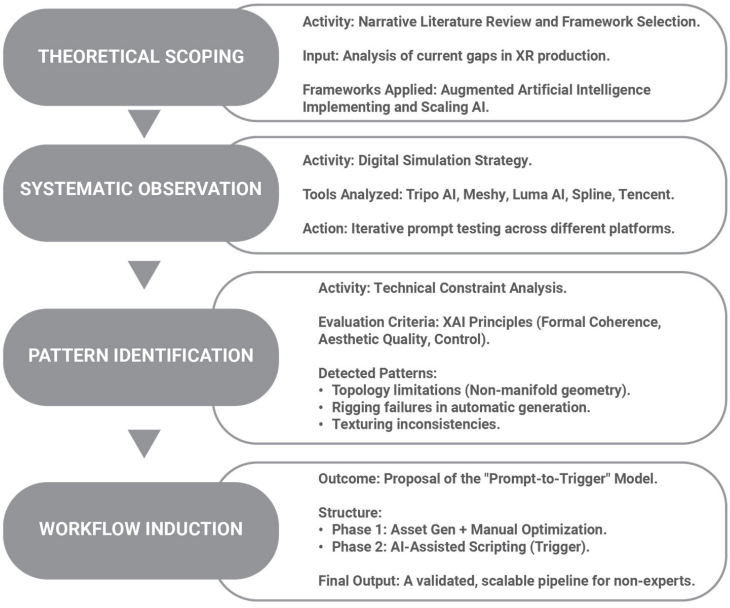


Figure 2. Structured methodological pipeline for framework development. Source: authors' elaboration.

To better illustrate the workflow, Table 1 shows how these criteria serve as an observational guide, focusing on iterative interactions to improve output quality and control, as well as interoperability with standard 3D/XR formats and pipelines. The criteria were defined following the multi-stage workflow and grounded in Explainable AI (XAI) principles [53], assessing prompt–output coherence, esthetic quality, and degree of creative control. This analysis makes it possible to identify specific technical limitations, such as topology errors that require manual intervention.

3.3. Phase 1: Documentation and Production Exploration

This methodological approach offers a thoughtful systematization of the use of GenAI tools in applied environments, which can reduce reliance on specialized technical skills and manual production costs [54], prioritizing a workflow accessible to non-specialized users and the documentation of practical design decisions. In Phase 1, the workflow serves as a space for visual ideation, where visual prompts and iterations help define the esthetic direction and generate resources for further development.

In Phase 1, the workflow supports visual ideation, where prompts and rapid visual iterations help define the esthetic direction and generate assets for development in subsequent phases. LLMs facilitate prompt formulation, iterative refinement, and step-by-step documentation. In parallel, image-generation tools support the creation of moodboards and style exploration. According to Kopiler et al. [55] these tools have accelerated the democratization of computer graphics by enabling the creation of complex visual content with fewer hardware and software constraints. In this workflow, they function as an ideation layer rather than a separate set of utilities, helping the visual direction early to reduce ambiguity before technical stages of 3D production. Consistent with Serna-Bernal and Rubio-Tamayo [56], GenAI not only optimizes time but redefines how complex concepts are ideated and visually represented through prompts. This shift has a direct impact on accessibility, allowing non-specialist users to clarify the visual direction of a project before facing the technical complexity of 3D production.

This section is crucial for XR projects because it connects early esthetic decisions with later 3D production. In specific contexts, a 3D registration or volumetric scanning phase is incorporated, supported by tools such as Gaussian Splatting, Luma AI, or reconstruction models based on photogrammetry or NeRF. As noted by Serna-Bernal and Rubio-Tamayo [56], incorporating supplementary capture routes into the asset creation phase enables the capture of visual and geometric information from the real world, converting it into processable digital material within graphics engines.

Complementarily, Kopiler et al. [55] highlight that use of representations such as Gaussian Splatting in Virtual Reality (VR) environments enables new paradigms of spatial editing facilitating the interactive manipulation of 3D Gaussian kernels. This phase serves as a complementary rather than essential component of the general model; the decision to employ these tools depends on the project's specific objectives, useful for capturing physical reality but not representing a mandatory condition for 3D production.

3.3.1. Introduction to Prompting: Preparing Instructions for 3D Generation

The training process addressed the operational logic of multimodal generative AI distinguishing the following four main integration axes: conversational, image, video and 3D generation. This distinction aligns with the Multimodal Artificial Intelligence framework described by Patil [57], which emphasizes integrating text, image and audio to enhance applications across sectors. Recognizing these categories was essential, as each architecture requires a specific prompt structure to operate effectively, rather than reiterating the general conversational tools (such as ChatGPT, Claude, Gemini or DeepSeek) [58]. The focus shifted

to the specialized transformer-based architecture used for multimodal generation, with the objective of understanding their capacity to maintain contextual coherence and process complex instructions.

The prompt was defined as an instruction that includes context, role, tone, and output conditions, enabling participants to apply the "working backwards" strategy—a method that anticipates intended outcomes to refine instructions. Although it is commonly used in corporate environments such as Amazon [59], this strategy has also been effectively adapted to support creative and design-oriented processes [60]. This logic is consistent with automated production workflows where textual source documents strictly dictate the visual output, reducing the need for manual iteration [54].

Ultimately, the goal was to develop textual strategies specifically for 3D environments, operating on the hypothesis that mastering the prompt is the new fundamental literacy for this workflow; as Serna-Bernal and Rubio-Tamayo [56] argue, this language directs the algorithmic generation of complex assets.

3.3.2. Preliminary Visualization of 3D Design via Image and Video Generation

Understanding these modifiers was essential because GenAI-assisted visual documentation supports pre-visualization, saves time, and reduces ambiguity before 3D modeling. This logic is consistent with automated production workflows in which textual guides visual outputs [54]. Building on this visual basis, the training then examined GenAI video-generation tools such as Runway, Kling AI, Pika, Sora, Genmo, and Veo 3. It should be noted that, of these, only Sora is referenced in the source documents as an emerging tool in the LLM ecosystem [61] and as a text-to-video model launched in February 2024 [62]. These tools allow transforming textual descriptions or images into audiovisual motion [63]; for example, video clips can be generated from the description of the desired content (text to video) or by applying a style and composition to a source video (video to video) [58].

Having identified video tools, it was explained that although these tools share elements with image generators, the prompt structure in video requires a more sequential narrative, describing not only the visual content but also the rhythm, camera movements, character actions, and approximate clip duration. Because of this, prompts are constructed following basic recommendations and elements to improve quality and accelerate the result.

Subsequently, the current limits of these tools were discussed, as a crucial step to understanding their viability in the workflow, where the difficulty in controlling the internal logic of movement or errors in the physics of objects or characters was identified [47]. This observation is supported by recent empirical studies by Motamed et al. [64], which demonstrates that current generative video models often fail to understand basic physical principles like gravity or solid mechanics. These limitations are important because GenAI is prone to errors and to generating false information, especially on complex topics. For this reason, the block helped clarify how visual design, narrative intention, and textual description must be articulated carefully to obtain coherent and visually effective outputs.

3.3.3. Texturing, Rigging and Technical Analysis

Firstly, texturing was analyzed, where some tools like Tripo AI, Meshy, or Scenario.gg include functions to apply materials, color, or textures; sometimes, UV mapping is embedded in the file (.glb or .fbx). However, it is not always possible to edit the maps in professional software, as many of these models present generic, redundant, or poorly adjusted textures relative to esthetic intent. This limitation aligns with findings in CAD workflows, where AI models show deficiencies in "UV layout design and texture mapping" [65]; they also tend to lose detail on curved or non-parametric surfaces.

After analyzing the texturing limitations, the rigging process continued. Although rigging is usually manual, tools like Tripo AI include basic automatic rigging that allows generating models ready for simple actions. Platforms like Mixamo (web platform; Adobe, San Jose, CA, USA; accessed on 21 July 2025) also support importing an .fbx model and applying predefined skeletons with associated animations.

Not all GenAI models were compatible with Mixamo, mainly because of topology errors, incorrect scaling, or non-uniform geometry. When the import worked correctly, animations such as idle, walk, run, or jump could be applied with minimal adjustments. Another persistent problem in the use of these tools is that automatic rigging generates rigid poses or artifacts in poorly articulated areas such as shoulders or hips, specifically when the model did not follow humanoid proportions. This suggests that geometry infidelity is often found in large model generations, reinforcing the need for a human-in-the-loop approach for iterative assessment and refinement [66].

GenAI 3D tools" because it is consistent with the focus of the article. Please revise the sentence as follows: "Once the rigs were applied, their functionality was tested using GenAI 3D tools such as Tripo AI, together with Mixamo, both used as web-based platforms, and Blender 4.2, where automatic skeletons were combined with the timeline to export animated sequences. In some cases, these animated models were also imported into Unity 2022.3.62f3 for testing in interactive environments. These tests served to verify if the generated model could be integrated without having to be rebuilt from scratch; in some cases, it was possible to import these animated models into Unity to perform tests in interactive environments.

The analysis of this phase concluded that, while IAGen 3D tools offer visually effective results, they still present limitations for seamless professional integration. Therefore, the models lack a logical object hierarchy and automatic rigging is typically non-editable. Consistent with Mondello et al. [67], who warn that workflows replacing human process entirely with AI tools can result in quality degradation, post-editing remains necessary to validate outputs in professional contexts; software such as Blender is required to validate these models.

3.3.4. Generation and Evaluation of 3D Models

All these previous blocks served as a necessary foundation for understanding the particularities of GenAI applied to 3D modeling. While each type of tool (conversational, image, and video) offers a key component, it was in the 3D generation block where these skills converged to give rise to a more complex workflow. Within the framework, the creation of three-dimensional models requires multiple levels of control. For that reason, it is indispensable to have previously understood how to structure an effective prompt and how to interpret the outputs. Prior work also indicates that 3D generation can rely on both text and visual references [68]; this aligns with Cheng et al. [69], whose work on SDFusion demonstrates that interactive generation must support a variety of input modalities, including images and text, to be truly functional within a creative pipeline.

During the modeling process and to test the prompt hypothesis, different types of input were experimented with. These included the simple textual, the composite prompt consisting of stylistic modifiers and technical details. This stepwise approach mirrors the pipeline described by Jun and Nichol [12], where a single synthetic view is first generated to guide the subsequent 3D structure. We also tested using a reference image as a sketch to improve structural coherence. Across this condition, the text + image configuration yielded the most consistent outputs in terms of visual control and formal precision. This superiority is technically grounded; as Liu et al. [70] argue, 3D reconstruction requires both "low-level perception" (provided by the image) and "high-level understanding" (provided by the

text), bridging the gap left by text-only descriptions. Consequently, the result depends not only on the prompt's detail but on the clarity of the intention and visual coherence.

Having analyzed the different inputs, the next logical step was to define practical criteria for assessing the generated 3D outputs. For this purpose, the models that were generated and evaluated using the following four criteria:

1. Mesh Quality: Focusing on topology and errors. As defined by Sorgente et al. [10], the goal is to produce a mesh with good geometrical properties within a target accuracy range for downstream applications.
2. Prompt Coherence: Comparing the input with the result to identify semantic deviations, including automated consistency metrics between rendered images and the input caption, similar to the CLIP R-Precision method evaluated by Poole et al. [71].
3. Visual Expressiveness: Assessing materials and textures. This addresses the challenge that almost no research has been conducted on the perceptual quality of texture-mapped models, as noted by Guo et al. [72], making subjective assessment critical.
4. Exportability: Verifying compatibility (formats .glb, .fbx, and .obj) and functionality in engines such as Unity, in line with the performance comparison of standard web formats established by Lee et al. [73].

The application of these four criteria highlighted recurring issues that define the central problem of this workflow, where the tools present limitations for fluid professional integration. Specifically, problems like non-editable geometries and generic texturing often result in assets that look rough or lack fine-detailed textures, confirming the observations of Jun and Nichol [12] regarding conditional implicit functions.

As a direct response to the limitations, the next step is technical processing; this is understood as the set of operations necessary to prepare the 3D object for integration, which involve complex surface cutting and parameter assignment to obtain semantically meaningful textures [11]. These elements transform the asset from a static model into a functional component. The result of Phase 1 is a selected set of assets and functional suggestions, along with the identification of their current limitations, which directly motivates the technical processing and optimization addressed in Phase 2.

3.4. Phase 2: Integration and Interaction Tests from Asset to Trigger

Building on the exploratory outputs of Phase 1, Phase 2 shifts from generation to production readiness by refining assets, optimizing performance, and integrating them into an XR prototype, including the implementation of trigger-based interactivity. The technical limitations and the need for substantial cleaning, identified in Phase 1, justify this second phase. The workflow begins with the generated 3D asset, which leads to the step of Manual Processing and Optimization. This clean-up stage is an indispensable methodological step. As Djaouti et al. [74] indicate the execution of logical rules depends on the precise detection of events such as collision to trigger state changes, without well-defined object boundaries, trigger-action logic becomes unreliable or may fail altogether.

This validates the findings of Chen et al. [75], who warn that "system performance" is a critical factor directly affecting user experience in virtual environments. Accordingly, sensory and performance-sensitive elements should be optimized before integration; for instance, Kawka et al. [25], mentioning the 3DforXR platform, emphasize that the objective is to generate optimized models ready for XR applications, highlighting that model size reduction and the generation of the "Level of Detail" (LOD) are crucial for improving performance in XR applications, which support the need for this processing step.

Once processed, the asset moves to platform integration; to maintain the workflow's accessibility focus, platform selection is grounded in the principle of "Democratizing Interactivity" [55]. Consistent with Rubio- Tamayo [76] who identifies "A-Frame is an ideal prototyping tool for the design of virtual spaces", and similarly, he identifies the Godot Game Engine as a key open source alternative, these tools facilitate access to content development without requiring industrial-grade hardware. However, the platform merely provides the environment; it does not inherently endow the 3D asset with behavior. To bridge the gap between the static environment and the desired interactivity, a structured logic is required.

To systematize how interactivity is constructed within these engines and to improve conceptual clarity, establish a functional differentiation of the Prompt-to- Trigger model based on the Event Condition Action (ECA) paradigm by Ceri and Ramakrishnan [77]:

- The prompt (Intent): Conceptualized as the input boundary where the user specifies the desired behavior [78]. It represents the human intent before computational processing.
- The asset (Entity): Refers to the sensory-rich virtual elements [75]. As noted by Djoauti et al. [74], these are passive entities (geometry/texture) that populate the environment but lack inherent agency.
- The trigger (Logic): Defined as the procedural middleware or "reactive rule" [74]. It is the logic unit that validates a condition (If) to execute an action (Then), effectively assigning agency to the static asset.

With the theoretical distinction between asset and logic established, the workflow proceeds to execution. In this phase, GenAI is utilized not to create the object itself, but to synthesize the logic script that governs its behavior. This approach substantiates [79] finding that machine learning streamlines production by automating metadata and logic generation. Within our proposed pipeline, LLMs are consulted to produce specific code structures (such as GDScript or HTML components), a strategy that aligns with Iacono et al. [80] and Constantinescu et al. [33]. These authors advocate for using AI to bridge the technical gap between novice and advanced creators by resolving implementation complexity.

Final interactivity result, the successful compilation of these scripts, leads to the ultimate objective: achieving a functional component capable of performing interactions standard to XR research. As explored by Gomez Maureira and Kniestedt [81], these interactive mechanics can range from "Reflex" actions (requiring fast reactions) to "Exploration" tasks involving the resolution of visual obstructions. Furthermore, Ur et al. [78] validate the necessity of "Remote Control" triggers, enabling direct user agency through natural methods such as voice commands or gestures, which aligns with the multimodal interactions described by [24]. By using AI to encode these behaviors from "spatial manipulation and proximity detection" as pointed out by Rubio-Tamayo et al. [76], to complex regulatory loops, the cycle is closed. This reflects the capacity to transform "declarative knowledge" (the prompt) into "reactive computations" (the trigger), a core function of active rules defined by Ceri and Ramakrishnan [77].

4. Findings and Observations

This section reports the empirical findings derived from the exploratory evaluation of GenAI-generated 3D assets and their integration constraints. These findings synthesize recurring technical issues (e.g., mesh editability, topology and export compatibility) that affect downstream use in real-time pipelines. The workflow models proposed in response to these findings are presented in the subsequent section (model proposal), where the Prompt-to-Trigger framework is articulated through Figures 3–5.

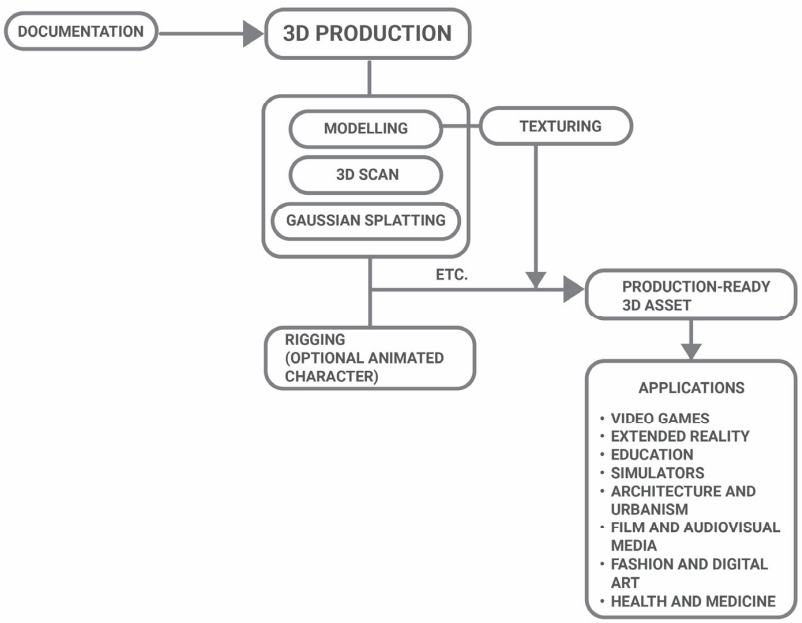


Figure 3. Applied Exploration Model. Source: authors' elaboration.

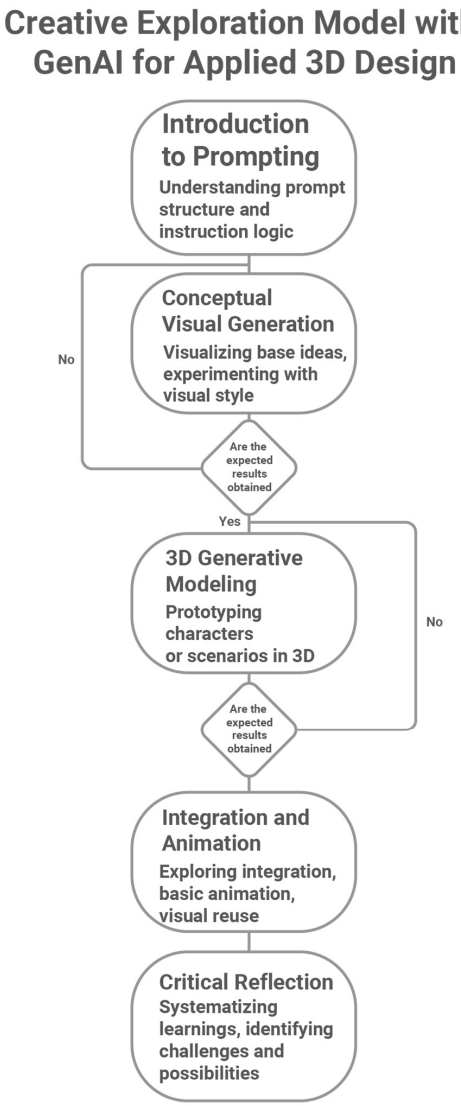


Figure 4. Creative Exploration Model with GenAI for applied 3D design. Source: authors' elaboration.

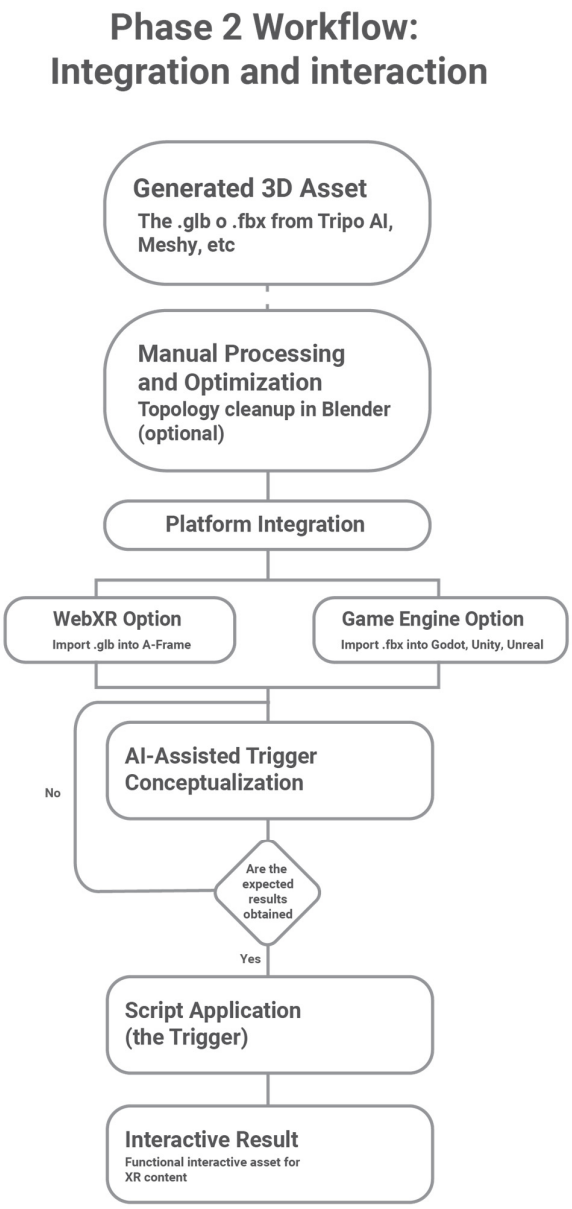


Figure 5. Asset to trigger workflow model. Source: authors' elaboration.

4.1. Evaluation Criteria

To make the exploratory findings transparent and comparable, generated 3D models were assessed using the following four criteria derived from common downstream requirements in real-time pipelines: (1) mesh quality (topology integrity, visible errors, and editability in Blender), (2) prompt coherence (alignment between the input prompt and the resulting form, identifying semantic drift), (3) level of detail and visual expressiveness (materials/textures, proportions, and perceived fidelity), and (4) exportability and compatibility (successful export and basic import checks using .glb/.gltf, .fbx, and .obj in tools such as Blender and real-time engines). Each criterion was rated qualitatively as Pass, Minor issues, Major issues, or Fail, and the observed issues were used to synthesize recurring limitations reported in the following subsection.

4.2. Findings from Phase 1: Limitations of Direct Production

The exploratory analysis of GenAI tools, platforms such as Meshy and Tripo AI, indicated that text-only prompting can yield acceptable outputs, but visual references generally improve results. When the instructions were ambiguous, the output reduced its

uniqueness and control over the asset. Regarding topology, the results contained mesh defects. In all the tools, the models generated with Meshy and Tencent showed better integration, while Tripo AI stood out for its simplicity.

These models still fall short of professional standards, as they cannot be edited directly during generations, forcing an iterative trial-and-error process. In some cases, the models were not recognized by Mixamo. According to this exploratory phase, it is demonstrated that GenAI tools do not deliver a final product, but rather a material that requires technical cleaning. These limitations are not the result, but rather validate the need for the structured workflow, connecting directly with the proposal of Phase 2.

5. Model Proposal: Prompt-to-Trigger Workflow

Existing work often treats GenAI 3D generation, asset optimization, and interaction scripting as separate problems. For non-expert users, treating these steps separately makes it hard to turn a good-looking generated asset into an XR prototype that works and responds to user input. The Prompt-to-Trigger model addresses this by organizing the process as one continuous workflow and by clarifying what happens between stages from GenAI ideation to generation, manual clean-up and optimization, to platform integration, and finally to trigger-based interaction. In doing so, the model addresses the accessibility gap by providing a practical pathway for non-expert users to prototype interactive XR experiences across applied domains.

5.1. Model 1 and Model 2: Production Models

As a response to the limitations of Phase 1, the models in Figures 4 and 5 are proposed, designed to guide non-expert users (e.g., designers, content creators, and practitioners) in the production process. The analysis of Phase 1 focused on several key tools, whose main characteristics are summarized below:

Generation Tools (Text/Image to 3D)

- Tripo AI, Meshy, Luma AI, Spline: these allow transforming natural language descriptions or images into basic 3D models, whether characters or objects, which stand out for their accessibility and speed for prototyping.
- Tencent ARC lab, Mirage, Scenario.gg: these are more specialized platforms, oriented towards stylized results, with greater esthetic control for the video game industry.
- Specific Functionalities Identified:
- Coherence Improvement (Meshy, Tripo AI): these allow uploading 2D images, which can be sketches or silhouettes, to guide the 3D generation and improve visual coherence.
- Automatic Animation (Tripo AI): it offers basic automatic rigging functions and exports with simple animations, representing a significant advance for non-experts.

Despite these functions, the platforms still require manual technical decisions; prompt quality remains a primary driver of output quality. In terms of application, these platforms are primarily used for visual prototyping, esthetic exploration, or rapid conceptual visualization in early stages of multimedia or educational projects.

The Applied Exploration Model (Figure 3) systematizes the operative core of 3D production. It starts with documentation; from there, the workflow enters the 3D production block, which groups the main routes through which a 3D asset can be created (modeling, 3D scan and Gaussian splatting); the routes represent alternative starting points that can be selected depending on the project needs. Texturing contributes to visual coherence and is applied as part of the production pipeline regardless; rigging is included as an optional stage, since it applies primarily to animated characters. The model also includes additional production tasks indicated by the etc. connector, which refers to common steps such as

mesh cleanup, UV adjustments, optimization, and export settings. These steps converge in a production-ready 3D asset, as outlined in Figure 3.

The Creative Exploration Model (Figure 4) captures how the process advances in practice through iterative checks. This feedback loop ensures the quality of each stage. It begins with the Introduction to Prompting, moves to Conceptual Visual Generation, and only if the quality targets are met does it advance to the Generative 3D Model. This iterative validation loop model is crucial so that a non-expert can systematize learning and recognize challenges before the integration phase.

5.2. Model 3: Asset-to-Trigger Workflow

This is the central model that connects Phase 1 with Phase 2 and resolves the problem of technical cleaning. The asset-to-trigger model (Figure 5) details the second half of the workflow.

1. The process starts with the generated 3D asset, which, as shown in Phase 1, often presents technical limitations.
2. This leads to the step of Manual Processing and Optimization, where any necessary clean-up is performed as needed in software such as Blender.
3. Once optimized the asset is integrated into a platform, prioritizing accessible environments such as WebXR (A-Frame) or a game engine (Godot, Unity, and Unreal).
4. The next stage is the Conceptualization of the AI-Assisted Trigger, where the designer's role acts as mediator using an LLM to generate script interaction.
5. The model includes a validation loop (Are expected results obtained?). If no, the conceptualization is revised and iterated; if yes, the application of the script (the trigger) is implemented.
6. The workflow ends with the Interactive Result: a functional asset for XR application.

These steps were applied in the single-asset worked example (Figure 6) which documents the Phase 1 and Phase 2 handoff from a GenAI draft to Blender-based post-processing and basic animation readiness check prior to XR integration.

These blocks show how the proposed workflow responds to the identified barriers by structuring the transition from asset generation to integration and interaction. This discussion also supports the suitability of the selected platforms for non-experts. As Rubio-Tamayo [76] proposes, "A-Frame is an ideal prototyping tool, which enables access to environment development", and likewise, he points out that Godot, due to its open-source licenses, facilitates access to content development. A practical implication is that post-processing (Step 2) functions as a viable bridge to these accessible platforms, since Kawka et al. [25] emphasize that "reduction of model size and the generation of the level of detail are crucial for improving performance in XR applications".

At the trigger conceptualization stage, the contribution of GenAI is best framed as support for implementation and skill development among non-expert creators. This aligns with the vision of Yosun Chang [26] who describes their platform as a 3D creation suite "accessible to all," similar to Iacono et al. [80] who propose GenAI as a "key to closing the gap between beginner and advanced creators". In practice, this support can take the form of consulting LLMs, for stepwise guidance and code scaffolding, which reinforces the feasibility of generating interaction scripts (triggers) with human oversight.

Figure 5 summarizes the core contribution of Phase 2 as a structured handoff from a GenAI-generated draft to a functional interactive prototype. Rather than treating generation, optimization, and interaction as separate tasks, the model organizes them as a single, sequential pathway that makes the dependencies between steps explicit. In this representation, the asset corresponds to the exported 3D output obtained in Phase 1, while

the trigger denotes the minimal interaction logic that operationalizes behavior within the target runtime.

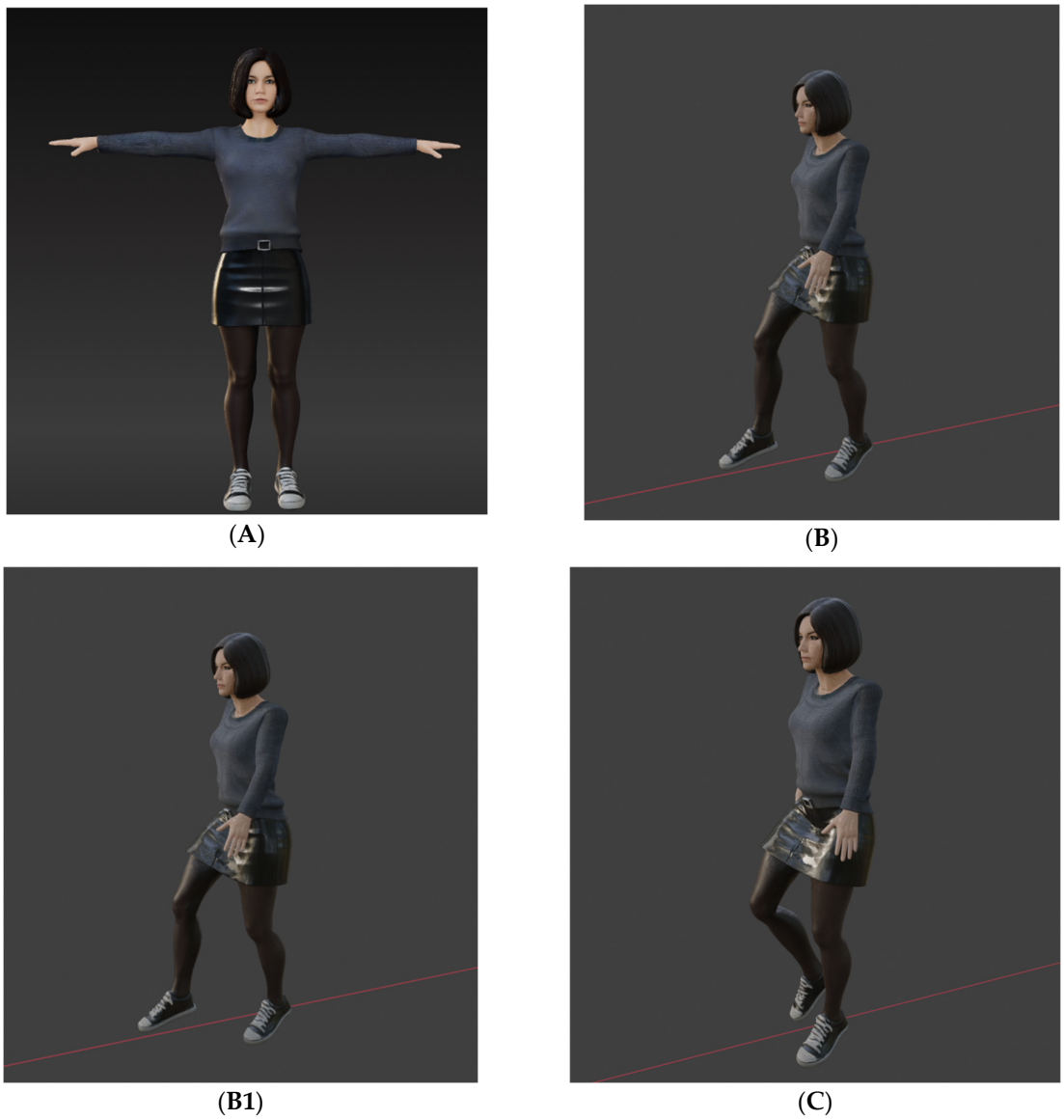


Figure 6. Single-asset example of the Phase 1–Phase 2 handoff in the Prompt-to-Trigger workflow: (**A**) GenAI draft asset; (**B**,**B1**) mesh inspection and cleanup in Blender, shown as a wireframe overlay and a detail view; (**C**) rigging and walk-cycle review and adjustment prior to XR deployment. Source: authors' elaboration.

5.3. Worked Example (Single Asset Demonstration)

To demonstrate how the workflow operates in practice, we present a single-asset worked example following the Phase 1–Phase 2 handoff. The example documents are the GenAI draft, Blender-based mesh inspection/clean-up, and a basic rigged walk animation as a readiness check prior to XR integration.

6. Discussion

To address the accessibility gap outlined in the Introduction, this section connects the proposed Prompt-to-Trigger workflow with the study's objective: to provide a structured path that can help non-expert creators move from asset generation to integration and interaction.

Phase 1 is presented as an exploratory stage focused on rapid prototyping, while Phase 2 covers the steps to prepare for production to generate XR assets with trigger-based logic. The observations from both phases clarify the role of each. Phase 1 shows that IAGen's current 3D tools are not consistent for final production, as objects may have limitations that affect their subsequent use in XR, highlighting the need for further cleanup and optimization when preparing assets for professional or real-time environments.

Phase 2 suggests that many of the limitations identified in Phase 1 can be addressed by non-expert users when the workflow incorporates a structured clean-up and optimization step. Substantial asset clean-up and optimization—identified by Kawka et al. [25] "crucial for improving performance"—acts as a practical bridge toward deployment. In addition, the use of accessible platforms such as A-Frame and Godot, which Rubio-Tamayo [76] describes as facilitating access to content development, can reduce barriers to entry for non-specialist creators in applied contexts. The workflow highlights GenAI's role in supporting the generation of basic interactions, even with limited implementation knowledge.

Figure 7 summarizes how the two phases connect in a comprehensive path that inexperienced users can follow in practical environments. Beyond this exploratory study, future evaluation should include usability/UX assessments, machine learning studies, and technical performance testing to evaluate the workflow under real-world user constraints, In XR, user-experience validation is commonly supported by standardized usability instruments because they are reliable and comparable across the systems. The System Usability Scale (SUS), in particular, is widely used in VR/AR/WebXR usability studies often alongside task-based measures and structured user feedback [82–85]. More recent evaluation work also stresses that XR validation should be treated as multidimensional, combining subjective UX evidence with objective indicators, and basic technical performance checks to relate perceived usability to deployability constraints [86,87].

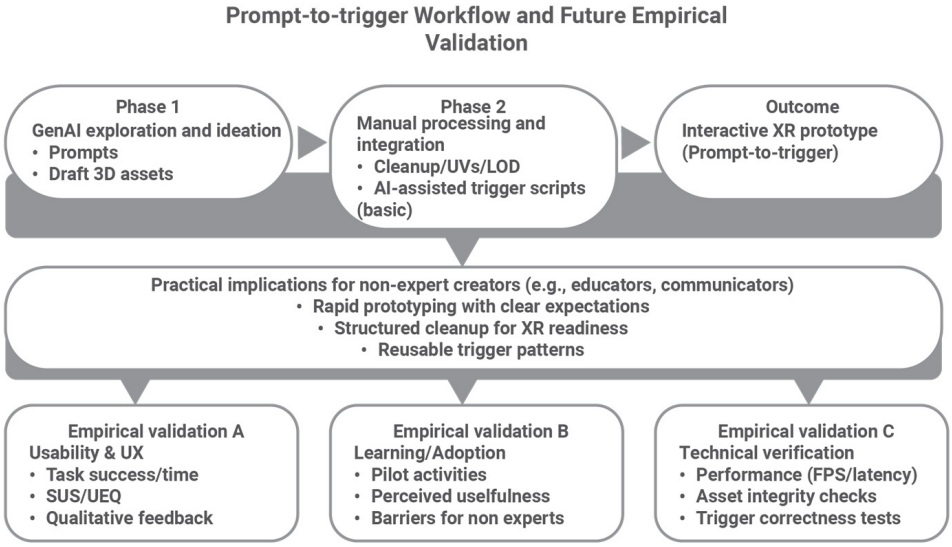


Figure 7. Overview of the Prompt-to-Trigger workflow (two phases) and evaluation pathways. Source: authors' elaboration.

As a basis for user-based validation, the SUS is proposed after participants complete a small set of representative tasks, aligned with both phases of the workflow. To maintain the design's scalability, the SUS can be supplemented with lightweight task measures that link perceived usability to connect user feedback with the practical results of the integrations. Table 2 summarizes these complementary evaluation pathways.

Table 2. Scalable evaluation design for future validation.

Pathway	Measures	Phase Mapping
Usability/UX (baseline)	SUS + brief feedback	Phase 1 + Phase 2
Task-based feasibility	Completion, time, error points	Phase 1 + Phase 2
Learning/adoption (optional)	Perceived usefulness/intention-to-use	Whole workflow
Technical performance	Import success, compatibility notes, runtime stability	Phase 2

The table describes a proposed evaluation plan for future work; no data from non-expert users are reported in this manuscript.

7. Conclusions

Applying the proposed methodology showed that image-guided prompting usually produces more coherent 3D drafts than text-only inputs. However, outputs across tools were rarely production-ready and often required manual clean-up due to recurring topology and texturing problems and, in some cases, rigging or animation issues. After this optimization step, the refined assets could be integrated into an XR environment. At that point, GenAI support remained useful as a practical aid for iterating trigger logic, helping non-expert users prototype basic interactive behaviors.

This article proposes the Prompt-to-Trigger workflow as a structured methodological model for creating XR prototypes with GenAI. In line with the study objectives and within the scope of this qualitative exploratory work, the workflow supports a complete path from prompt-based ideation and draft asset generation in Phase 1 to deployment-oriented refinement and basic trigger-based interactivity in Phase 2, resulting in an integrated interactive XR prototype. Compared with approaches that treat generation, optimization, or scripting as separate tasks, the main contribution is an end-to-end framing that makes the handoffs between stages explicit and helps teams replicate the prototyping process across applied contexts, including communication and education.

Regarding the study hypotheses, the findings support them within the limits of this exploratory approach. H1 is supported because current GenAI tools enable non-expert creators to generate initial 3D drafts, especially when prompts are guided by images. H2 is supported because, after manual optimization, assets could be integrated into an XR environment. H3 is supported because AI-assisted scripting helped reduce entry barriers when building and iterating trigger logic, although human supervision was still needed for debugging and deployment constraints. Future work should evaluate the workflow with broader user studies and more systematic benchmarks to compare tools and measure outcomes across contexts, as outlined in Figure 7.

Limitations and Future Work

This study followed a qualitative, exploratory approach. Trigger conceptualization was limited to basic interaction rules and did not extend to the implementation of a complex NLP-driven interaction system. Future work should apply the workflow in empirical case studies with non-expert users and include quantitative performance analyses. Performance and integration tests of optimized assets should be compared across XR platforms, alongside usability and adoption measures to assess the workflow under realistic production constraints.

Author Contributions: Conceptualization, J.L.R.T. and M.A.S.B.; methodology, J.L.R.T. and M.A.S.B.; formal analysis, J.L.R.T. and M.A.S.B.; investigation, J.L.R.T. and M.A.S.B.; writing—original draft preparation, J.L.R.T. and M.A.S.B.; visualization, J.L.R.T. and M.A.S.B.; supervision, J.L.R.T.; project administration J.L.R.T. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: This document consists of a literature review and the proposal of a potentially scalable theoretical model that can be further refined and restructured. Through a series of diagrams, it presents the development phases of 3D assets for integration into extended reality environments and game engines, supported by generative AI. The model proposal, based on these workflow phases, is described visually across the diagrams included in the manuscript. In addition, Appendix A illustrates and specifies the different stages of a model intended to frame future research across the phases of a 3D asset creation and development process for extended reality. All relevant information, including the diagrams, is contained within the manuscript. Since this work proposes a theoretical model grounded in a literature review, there are no external raw data tables beyond the information provided in the manuscript.

Acknowledgments: This project has been developed as part of the activities of the Immersive and Interactive Communication Laboratory (XR COM LAB) at URJC, within the REDLABU network. The XR COM LAB is part of the Extended Reality Laboratory (XR Unit) of the CAT in Fuenlabrada, Madrid, Spain.

Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A. Evaluation Framework for the 'Prompt-to-Trigger' Workflow

This appendix provides a structured overview of how the proposed "Prompt-to-Trigger" workflow is established within the study's exploratory design. The diagram describes the procedural logic followed to assess the feasibility of the workflow across the main stages outlined: research design, prompt-based generation of 3D components, manual optimization of components and integration into XR environments, AI-assisted trigger iteration, and a final synthesis based on content typology.

Figure A1 summarizes this process and clarifies how it can be structured as an adaptable and scalable tool when designing standards and workflows to develop and evaluate generated 3D content and its integration into the XR environment.

In the first phase, the aim is to determine how current GenAI tools can help non-experts generate initial 3D sketches. Furthermore, we consider whether increased workflow hybridization can facilitate the process of integrating these items into extended reality (XR) environments following their optimization, and whether AI-assisted scripting can help reduce the technical barriers associated with trigger conceptualization.

Phase 2 focuses on preparing content for production, XR integration and interaction support. At this stage, the generated assets undergo a manual editing process involving post-processing and optimization. The aim is to improve their usability, as well as other factors such as suitability for the medium, within the contexts in which they are to be implemented. Among these factors, the main ones to highlight are the cleaning and optimization of meshes, the adjustment of textures and scale—which is not universal—and format compatibility, for the subsequent preparation of the items for integration into accessible XR platforms or game engines. Once integrated, AI assistance is proposed to support the creation of code structures and the conceptualization of triggers based on the design of the environment's interactions. This would allow the basic interaction logic to be iterated through a loop of expected results, which are then tested and refined. This phase highlights the fact that, following optimization, the generated assets can be integrated into XR environments or game engines, and that AI-assisted scripting can help prototype simple interactive behaviors to scale production to more complex generation models, whether in the realm of modeling, narrative, interaction development or environment design.

The final synthesis is based on the convergence of both phases. This set of conceptual models provides, in a clear, visual and structured manner, as well as in an adaptable and scalable format, a preliminary guide for future studies aiming to analyze, in detail or jointly, the phases of 3D production and their integration into XR environments, including variants that incorporate GenAI assistance. The complexity of the field requires this type of model to act as a map in which specific studies and tool choices can be situated.

Furthermore, the scalability and flexibility of these models are essential as other factors related to technological advancements in areas such as 3D production, integration into environments, frameworks and game engines for XR, and the use of GenAI as an assistive tool in different phases of the process are incorporated.

The aim of Appendix A is to strengthen the methodological understanding of the proposed model and to present, in a graphical format, the phases into which the model is structured, with a view to designing, in future work, workflow evaluation procedures in the field of 3D creation for extended reality, thereby offering an initial scalable model framework.

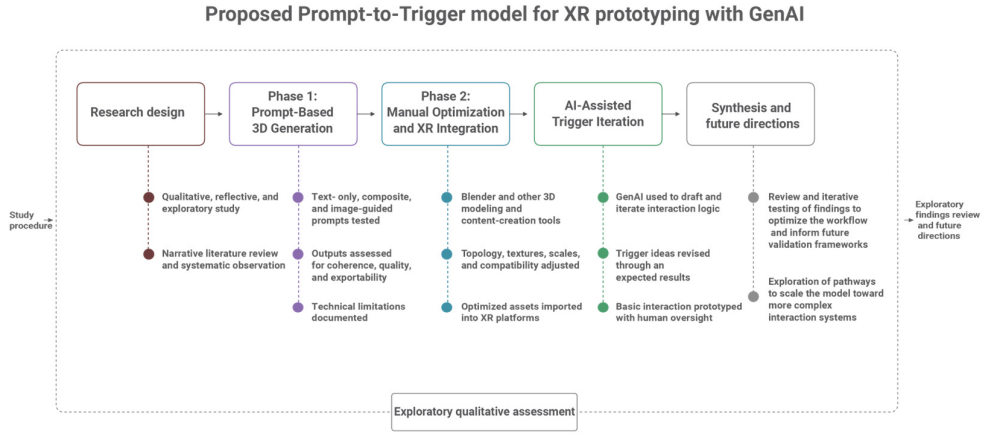


Figure A1. Diagram of the workflow development phases. The figure summarizes how the proposed workflow is examined through the exploratory design of the study, from the research design to the structuring of the different phases through a review of the literature and the identification of specific characteristics in the phases of 3D content production for extended reality.

References

1. Kretzschmar, M.; Dammann, M.P.; Schwoch, S.; Braun, F.; Saske, B. Evaluating the Current Role of Generative AI in Engineering Development and Design—A Systematic Review. In Proceedings of the DS 130: Proceedings of NordDesign 2024, Reykjavik, Iceland, 12–14 August 2024.
2. Abouelenein, Y.A.M.; Nagy Elmaadaway, M.A. Impact of Teaching a Neuro-Computerized Course Through VLE to Develop Computational Thinking Among Mathematics Pre-Service Teachers. *J. Educ. Comput. Res.* **2023**, *61*, 1175–1206. [CrossRef]
3. Lee, H. A Unified Conceptual Model of Immersive Experience in Extended Reality. *Comput. Hum. Behav. Rep.* **2025**, *18*, 100663. [CrossRef]
4. Ratican, J.; Hutson, J.; Wright, A. A Proposed Meta-Reality Immersive Development Pipeline: Generative AI Models and Extended Reality (XR) Content for the Metaverse. *J. Intell. Learn. Syst. Appl.* **2023**, *15*, 24–35. [CrossRef]
5. Gong, L.; Fast-Berglund, A.; Johansson, B. A Framework for Extended Reality System Development in Manufacturing. *IEEE Access* **2021**, *9*, 24796–24813. [CrossRef]
6. Gornall, J.; Peña, A.; Pinto, H.; Rojas, J.; Correa, F.; García, J. Extended Realityin Construction 4.0: A Systematic Review of Applications, Implementation Barriers, and Research Trends. *Appl. Sci.* **2025**, *16*, 9. [CrossRef]
7. Rivera, F.M.-L.; Mora-Serrano, J.; Oñate, E.; Montecinos-Orellana, S. A Comprehensive Framework for Integrating Extended Reality into Lifecycle-Based Construction Safety Management. *Appl. Sci.* **2025**, *15*, 5690. [CrossRef]
8. Engelbrecht, H.A.; Rossi, S.; Gunkel, S.N.B. *Proceedings of the 16th ACM Multimedia Systems Conference*; Association for Computing Machinery: New York, NY, USA, 2025; ISBN 979-8-4007-1467-2.
9. Rizvic, S.; Mijatovic, B.; Boskovic, D.; Ivkovic-Kihic, I. Workflow of Extended Reality Applications for Museum Exhibitions. In Proceedings of the 2022 International Balkan Conference on Communications and Networking (BalkanCom), Sarajevo, Bosnia and Herzegovina, 22–24 August 2022; pp. 189–194.

10. Sorgente, T.; Biasotti, S.; Manzini, G.; Spagnuolo, M. A Survey of Indicators for Mesh Quality Assessment. *Comput. Graph. Forum* **2023**, *42*, 461–483. [CrossRef]
11. Chen, Y.; Liu, X.; Li, Y.; Cheung, V.; Chen, Z.; Zhang, D.; Guo, C. ArtUV: Artist-Style UV Unwrapping. *arXiv* **2025**, arXiv:2509.20710.
12. Jun, H.; Nichol, A. Shap-E: Generating Conditional 3D Implicit Functions. *arXiv* **2023**, arXiv:2305.02463. [CrossRef]
13. Tu, X.; Autiosalo, J.; Ala-Laurinaho, R.; Yang, C.; Salminen, P.; Tammi, K. TwinXR: Method for Using Digital Twin Descriptions in Industrial eXtended Reality Applications. *Front. Virtual Real.* **2023**, *4*, 1019080. [CrossRef]
14. Qayyum, A.; Butt, M.A.; Ali, H.; Usman, M.; Halabi, O.; Al-Fuqaha, A.; Abbasi, Q.H.; Imran, M.A.; Qadir, J. Secure and Trustworthy Artificial Intelligence-Extended Reality (AI-XR) for Metaverses. *ACM Comput. Surv.* **2024**, *56*, 1–38. [CrossRef]
15. Memarian, B.; Doleck, T. A Novel Ethical Analysis of Educational XR and AI in Literature. *Comput. Educ. X Real.* **2024**, *4*, 100052. [CrossRef]
16. Alkaeed, M.; Qayyum, A.; Qadir, J. Privacy Preservation in Artificial Intelligence and Extended Reality (AI-XR) Metaverses: A Survey. *J. Netw. Comput. Appl.* **2024**, *231*, 103989. [CrossRef]
17. Guo, X.; Guo, Y.; Liu, Y. The Development of Extended Reality in Education: Inspiration from the Research Literature. *Sustainability* **2021**, *13*, 13776. [CrossRef]
18. Rangel-de Lázaro, G.; Duart, J.M. You Can Handle, You Can Teach It: Systematic Review on the Use of Extended Reality and Artificial Intelligence Technologies for Online Higher Education. *Sustainability* **2023**, *15*, 3507. [CrossRef]
19. Zwoliński, G.; Kamińska, D.; Laska-Leśniewicz, A.; Haamer, R.E.; Vairinhos, M.; Raposo, R.; Urem, F.; Reisinho, P. Extended Reality in Education and Training: Case Studies in Management Education. *Electronics* **2022**, *11*, 336. [CrossRef]
20. Barbu, M.; Iordache, D.; Petre, I.; Barbu, D.; Bajenaru, L. Framework Design for Reinforcing the Potential of XR Technologies in Transforming Inclusive Education. *Appl. Sci.* **2025**, *15*, 1484. [CrossRef]
21. Gu, Y.; Feijoo-Garcia, M.A.; Zhang, Y.; Magana, A.J.; Benes, B.; Popescu, V. An XR Environment for AI Education: Design and First Implementation. In Proceedings of the 2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW), Orlando, FL, USA, 16–21 March 2024; pp. 157–162.
22. Akhtar, Z.B.; Rawol, A.T. Artificial Intelligence (AI) and Extended Reality (XR): A Biomedical Engineering Perspective Investigation Analysis. *Indones. J. Electron. Electromed. Eng. Med. Inform.* **2024**, *6*, 132–146. [CrossRef]
23. Mande, P.B.; Bhaskar, R.A. AI-powered virtual reality and augmented reality for immersive learning in colleges. *Int. J. Adv. Sci. Res.* **2025**, *10*, 139–142.
24. Yan, W.; Li, B.; Lowell, V.L. Integrating Artificial Intelligence and Extended Reality in Language Education: A Systematic Literature Review (2017–2024). *Educ. Sci.* **2025**, *15*, 1066. [CrossRef]
25. Kawka, M.S.; Nikopoulos, A.; Stentoumis, C.; Kalisperakis, I. 3DforXR: An AI-Driven Multi-Input Pipeline for 3D Object Generation in XR Applications. In Proceedings of the 2024 IEEE 3rd International Conference on Intelligent Reality (ICIR), Coimbra, Portugal, 5–6 December 2024; pp. 1–4.
26. Chang, Y. AI3D Desktop: Creating 3D Using AI in Skeuomorphic XR and The Future of Work. In *Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Appy Hour, Denver, CO, USA, 27 July–1 August 2024*; ACM: New York, NY, USA, 2024; pp. 1–2.
27. Sahebnasi, M.J.; Farrokhimaleki, M.; Wang, N.; Zhao, R.; Maurer, F. Exploring the Potential of Generative AI in Prototyping XR Applications. In Proceedings of the RealXR: Prototyping and Developing Real-World Applications for Extended Reality, Arenzano, Italy, 4 June 2024.
28. Tamayo, J.L.R.; Levratto, V.; Gómez, H.G. Applications in 3D Volumetric Video for the Prototyping of Immersive and Interactive Enviroments for Education and Research: Prototyping and Component Integration In Extended Reality. In Proceedings of the 19th International Technology, Education and Development Conference, Valencia, Spain, 3–5 March, 2025; pp. 2544–2550.
29. Regassa Hunde, B.; Debebe Woldeyohannes, A. Future Prospects of Computer-Aided Design (CAD)—A Review from the Perspective of Artificial Intelligence (AI), Extended Reality, and 3D Printing. *Results Eng.* **2022**, *14*, 100478. [CrossRef]
30. Fakhimi, M.; Saadat, N. Integration of AI and XR in Optimizing Human-Centered Architectural Spaces: A Scoping Review. *J. Propuls. Technol.* **2024**, *45*, 3838–3866.
31. Vallasciani, G.; Stacchio, L.; Cascarano, P.; Marfia, G. CreAIXR: Fostering Creativity with Generative AI in XR Environments. In Proceedings of the 2024 IEEE International Conference on Metaverse Computing, Networking, and Applications (MetaCom), Hong Kong, China, 12–14 August 2024; pp. 1–8.
32. Sun, K. A New Era of Visual Storytelling: Integrating Generative AI and XR in Undergraduate Virtual Production Pedagogy. Available online: https://www.worldscientific.com/doi/epdf/10.1142/S0129156425409088 (accessed on 9 November 2025).
33. Constantinescu, G.-G. Emerging Trends and Computational Approaches in the Application of XR Technologies in Education. In *Proceedings of the International Conference IMCS-60, MSU, Chisinau, Republic of Moldova, 10–13 October 2024*; Vladimir Andrunachievici Institute of Mathematics and Computer Science, MSU: Chisinau, Republic of Moldova, 2024; pp. 276–286.

34. Mendoza Garcia, C.I.; Munoz Guzman, A.D.; Sanchez Ruano, D. Binding AI and XR in Design Education: Challenges and Oppotunities with Emerging Technologies. In *Proceedings of the International Conference on Engineering and Product Design Education, EPDE 2024*; The Design Society: Glasgow, UK, 2024; pp. 247–251.
35. Talaat, F.M.; Hassan, E. Artificial Intelligence in 3D Printing. In *Proceedings of the Enabling Machine Learning Applications in Data Science*; Hassanien, A.E., Darwish, A., Abd El-Kader, S.M., Alboaneen, D.A., Eds.; Springer: Singapore, 2021; pp. 77–88.
36. Guo, K.; Yang, Z.; Yu, C.-H.; Buehler, M.J. Artificial Intelligence and Machine Learning in Design of Mechanical Materials. *Mater. Horiz.* **2021**, *8*, 1153–1172. [CrossRef] [PubMed]
37. Choi, W.; Advincula, R.C.; Wu, H.F.; Jiang, Y. Artificial Intelligence and Machine Learning in the Design and Additive Manufacturing of Responsive Composites. *MRS Commun.* **2023**, *13*, 714–724. [CrossRef]
38. Charalampous, P.; Kladovasilakis, N.; Pelekoudas, T.; Kostavelis, I.; Ioannidis, D.; Tzovaras, D. Product Design of Customized 3D Printed Knives Using Artificial Intelligence and Numerical Simulation Technologies. *Procedia Comput. Sci.* **2025**, *253*, 1770–1779. [CrossRef]
39. Berg, L.P.; Vance, J.M. Industry Use of Virtual Reality in Product Design and Manufacturing: A Survey. *Virtual Real.* **2017**, *21*, 1–17. [CrossRef]
40. Berni, A.; Borgianni, Y. Applications of Virtual Reality in Engineering and Product Design: Why, What, How, When and Where. *Electronics* **2020**, *9*, 1064. [CrossRef]
41. Luck, M.; Aylett, R. Applying Artificial Intelligence to Virtual Reality: Intelligent Virtual Environments. *Appl. Artif. Intell.* **2000**, *14*, 3–32. [CrossRef]
42. Geraci, R.M. *Apocalyptic AI: Visions of Heaven in Robotics, Artificial Intelligence, and Virtual Reality*; Oxford University Press: New York, NY, USA, 2010; ISBN 978-0-19-997893-9.
43. Bertol, D. *Designing Digital Space: An Architect's Guide to Virtual Reality*; John Wiley & Sons: Hoboken, NJ, USA, 1996; ISBN 978-0-471-14662-9.
44. Wu, Y. Application of Artificial Intelligence within Virtual Reality for Production of Digital Media Art. *Comput. Intell. Neurosci.* **2022**, *2022*, 3781750. [CrossRef]
45. Gong, Y. Application of Virtual Reality Teaching Method and Artificial Intelligence Technology in Digital Media Art Creation. *Ecol. Inform.* **2021**, *63*, 101304. [CrossRef]
46. Peckham, O.; Raines, J.; Bulsink, E.; Goudswaard, M.; Gopsill, J.; Barton, D.; Nassehi, A.; Hicks, B. Artificial Intelligence in Generative Design: A Structured Review of Trends and Opportunities in Techniques and Applications. *Designs* **2025**, *9*, 79. [CrossRef]
47. Sakirin, T.; Kusuma, S. A Survey of Generative Artificial Intelligence Techniques. *Babylon. J. Artif. Intell.* **2023**, *2023*, 10–14. [CrossRef]
48. Li, J.; Cao, H.; Lin, L.; Hou, Y.; Zhu, R.; El Ali, A. User Experience Design Professionals' Perceptions of Generative Artificial Intelligence. In *Proceedings of the CHI Conference on Human Factors in Computing Systems, Honolulu, HI, USA, 11–16 May 2024*; ACM: New York, NY, USA, 2024; pp. 1–18.
49. Gorban, A.N.; Grechuk, B.; Tyukin, I.Y. Augmented Artificial Intelligence: A Conceptual Framework. *arXiv* **2018**, arXiv:1802.02172. [CrossRef]
50. Haefner, N.; Parida, V.; Gassmann, O.; Wincent, J. Implementing and Scaling Artificial Intelligence: A Review, Framework, and Research Agenda. *Technol. Forecast. Soc. Change* **2023**, *197*, 122878. [CrossRef]
51. Begemann, A.; Hutson, J. Empirical Insights into AI-Assisted Game Development: A Case Study on the Integration of Generative AI Tools in Creative Pipelines. *Metaverse* **2024**, *5*, 2568. [CrossRef]
52. Tang, Y.; Huang, Y.; Wang, H.; Wang, C.; Guo, Q.; Yao, W. Framework for Artificial Intelligence Analysis in Large-Scale Power Grids Based on Digital Simulation. *CSEE J. Power Energy Syst.* **2018**, *4*, 459–468. [CrossRef]
53. Angelov, P.P.; Soares, E.A.; Jiang, R.; Arnold, N.I.; Atkinson, P.M. Explainable Artificial Intelligence: An Analytical Review. *WIREs Data Min. Knowl. Discov.* **2021**, *11*, e1424. [CrossRef]
54. Dhanagopal, K. Automated Video Production Workflow Using Textual Source Documents. Available online: https://www.tdcommons.org/dpubs_series/8659/ (accessed on 1 January 2026).
55. Kopiler, A.A.; Schardong, G.; Schirmer, L.; Perazzo, D.; Novello, T.; Velho, L. Democratizing Interactivity: An Overview of Interfaces for Multimedia Machine Learning. *Comput. Graph.* **2025**, *132*, 104412. [CrossRef]
56. Serna-Bernal, M.A.; Rubio-Tamayo, J.-L. Inteligencia artificial generativa en procesos de diseño visual para entornos multimedia interactivos: Análisis de pipelines y flujos de trabajo y propuesta de modelo. *Hipertext.Net* **2025**, *31*, 11–24. [CrossRef]
57. Patil, D. *Multimodal Artificial Intelligence in Industry: Integrating Text, Image, and Audio for Enhanced Applications across Sectors*; SSRN: Rochester, NY, USA, 2024.
58. Banh, L.; Strobel, G. Generative Artificial Intelligence. *Electron. Mark.* **2023**, *33*, 63. [CrossRef]
59. Ates, A.; Suppayah, K. Disciplined Innovation: A Case Study of the Amazon Working Backwards Approach to Internal Corporate Venturing. *Res.-Technol. Manag.* **2024**, *67*, 23–33. [CrossRef]

60. Harisman, Y.; Ashari, J.; Amam, A.; Harun, L. The Analysis Application of the Working Backwards Strategy in Mathematic Problem Solving for Senior High School Students between Gender; San Diego, California (USA). *AIP Conf. Proc.* **2023**, *2698*, 060024.
61. Carbonell-Alcocer, A.; Sanchez-Acedo, A.; Benitez-Aranda, N.; Gertrudix, M. Impacto de la Inteligencia Artificial Generativa en la eficiencia, calidad e innovación en la producción de Recursos Educativos Abiertos para MOOCS. *Comun. Soc.* **2024**, *22*, 1–31. [CrossRef]
62. Bordas, A.; Le Masson, P.; Thomas, M.; Weil, B. What Is Generative in Generative Artificial Intelligence? A Design-Based Perspective. *Res. Eng. Des.* **2024**, *35*, 427–443. [CrossRef]
63. Agostinelli, A.; Denk, T.I.; Borsos, Z.; Engel, J.; Verzetti, M.; Caillon, A.; Huang, Q.; Jansen, A.; Roberts, A.; Tagliasacchi, M.; et al. MusicLM: Generating Music From Text. *arXiv* **2023**, arXiv:2301.11325. [CrossRef]
64. Motamed, S.; Culp, L.; Swersky, K.; Jaini, P.; Geirhos, R. Do Generative Video Models Understand Physical Principles? In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*; IEEE Computer Society: Washington, DC, USA, 2025.
65. Kung, C.-H.; Liang, K.-C. Exploring the Usability and Future Development of AI-Generated 3D Models in CAD Workflows and the Metaverse Based on 3D Model Standards. *Comput.-Aided Des. Appl.* **2025**, *22*, 782–804. [CrossRef]
66. Sadik, A.R.; Bujny, M. Human-in-the-Loop: Quantitative Evaluation of 3D Models Generation by Large Language Models. *arXiv* **2025**, arXiv:2509.07010.
67. Mondello, A.; Cini, R.; Rasane, S.; Karakanta, A.; Casanellas, L. Using AI Tools in Multimedia Localization Workflows: A Productivity Evaluation. In *Proceedings of Machine Translation Summit XX: Volume 2*; European Association for Machine Translation: Gentbrugge, Belgium, 2025.
68. Lin, C.-H.; Gao, J.; Tang, L.; Takikawa, T.; Zeng, X.; Huang, X.; Kreis, K.; Fidler, S.; Liu, M.-Y.; Lin, T.-Y. Magic3D: High-Resolution Text-to-3D Content Creation. In Proceedings of the 2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Vancouver, BC, Canada, 17–24 June 2023.
69. Cheng, Y.-C.; Lee, H.-Y.; Tulyakov, S.; Schwing, A.; Gui, L. SDFusion: Multimodal 3D Shape Completion, Reconstruction, and Generation. In *Proceedings of the 2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Vancouver, BC, Canada, 17–24 June 2023*; IEEE: Piscataway, NJ, USA, 2023; pp. 4456–4465.
70. Liu, R.; Wu, R.; Hoorick, B.V.; Tokmakov, P.; Zakharov, S.; Vondrick, C. Zero-1-to-3: Zero-Shot One Image to 3D Object. In Proceedings of the 2023 IEEE/CVF International Conference on Computer Vision (ICCV), Paris, France, 1–6 October 2023.
71. Poole, B.; Jain, A.; Barron, J.T.; Mildenhall, B. DreamFusion: Text-to-3D Using 2D Diffusion. *arXiv* **2022**, arXiv:2209.14988.
72. Guo, J.; Vidal, V.; Cheng, I.; Basu, A.; Baskurt, A.; Lavoue, G. Subjective and Objective Visual Quality Assessment of Textured 3D Meshes. *ACM Trans. Appl. Percept.* **2017**, *14*, 1–20. [CrossRef]
73. Lee, J.; Eom, S.-Y.; Lee, J. Empowering Game Designers with Generative AI. *IADIS Int. J. Comput. Sci. Inf. Syst.* **2023**, *18*, 213–230. [CrossRef]
74. Djaouti, D.; Alvarez, J.; Jessel, J.-P.; Methel, G.; Molinier, P. A Gameplay Definition through Videogame Classification. *Int. J. Comput. Games Technol.* **2008**, *2008*, 470350. [CrossRef]
75. Chen, M.-X.; Hu, H.; Yao, R.; Qiu, L.; Li, D. A Survey on the Design of Virtual Reality Interaction Interfaces. *Sensors* **2024**, *24*, 6204. [CrossRef] [PubMed]
76. Rubio-Tamayo, J.L. *A-Frame as a Prototyping Tool for the Design of Immersive Experiences in Cultural Heritage: Towards an Extended Reality Approach to Interaction Design*; CITCEM: Porto, Portugal, 2024; ISBN 978-989-8970-92-3.
77. Ceri, S.; Ramakrishnan, R. Rules in Database Systems. *ACM Comput. Surv.* **1996**, *28*, 109–111. [CrossRef]
78. Ur, B.; McManus, E.; Pak Yong Ho, M.; Littman, M.L. Practical Trigger-Action Programming in the Smart Home. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems, Toronto, ON, Canada, 26 April–1 May 2014*; Association for Computing Machinery: New York, NY, USA, 2014; pp. 803–812.
79. Steele, K. Streamlining Production Workflows for Accessible Audio Using AI Metadata Assignment. Ph.D. Thesis, University of York, York, UK, 2023.
80. Iacono, S.; Zolezzi, D.; Vercelli, G. Metaverse, A Year After: Evolution of XR Tools and Generative-AI. In Proceedings of the 2023 IEEE International Conference on Metrology for eXtended Reality, Artificial Intelligence and Neural Engineering (MetroXRAINE), Milano, Italy, 25–27 October 2023; pp. 34–39.
81. Gómez-Maureira, M.A.; Kniestedt, I. Exploring Video Games That Invoke Curiosity. *Entertain. Comput.* **2019**, *32*, 100320. [CrossRef]
82. Arpaia, P.; De Benedetto, E.; De Paolis, L.; D'Errico, G.; Donato, N.; Duraccio, L. Performance and Usability Evaluation of an Extended Reality Platform to Monitor Patient's Health during Surgical Procedures. *Sensors* **2022**, *22*, 3908. [CrossRef]
83. Kim, H.-Y. Development and Usability Assessment of Virtual Reality- and Haptic Technology-Based Educational Content for Perioperative Nursing Education. *Healthcare* **2024**, *12*, 1947. [CrossRef]
84. Dutta, R.; Mantri, A.; Singh, G. Evaluating System Usability of Mobile Augmented Reality Application for Teaching Karnaugh-Maps. *Smart Learn. Environ.* **2022**, *9*, 6. [CrossRef]

85. Rzeszewski, M.; Orylski, M. Usability of WebXR Visualizations in Urban Planning. *ISPRS Int. J. Geo-Inf.* **2021**, *10*, 721. [CrossRef]
86. Salinas, D.; Rivera, F.M.-L.; Mora-Serrano, J. Critical Analysis of the Evaluation Methods of Extended Reality (XR) Experiences for Construction Safety. *Int. J. Environ. Res. Public Health* **2022**, *19*, 15272. [CrossRef]
87. Picardi, A.; Caruso, G. User-Centered Evaluation Framework to Support the Interaction Design for Augmented Reality Applications. *Multimodal Technol. Interact.* **2024**, *8*, 41. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.